SUPERVISED MACHINE LEARNING

HOUSE PRICE PREDICTION

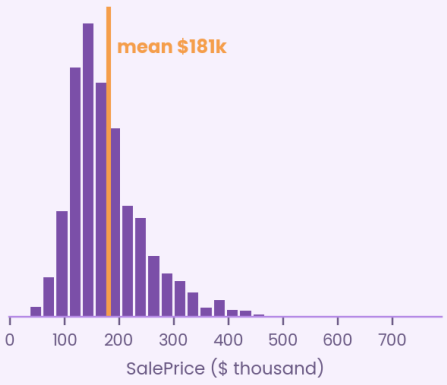
Step 1: Problem Framing & Theory Notes

Regression Analysis on the Ames Housing Dataset

Simple, Multiple, Ridge and Lasso regression for predicting house sale prices from 79 features.

Sale price distribution

1,460 houses in the Ames dataset



Why it matters

cost of a wrong price prediction

Under-prediction — seller loses

Over-prediction — unsold house

goal: keep the error (RMSE) small

1,460
houses in train.csv

$180,921
average sale price

79
input features

01 Frame
Regression vs classification

02 Evaluate
RMSE, MAE, R^2 & k-fold CV

03 Choose
Ridge / Lasso & λ

Step 1: Problem Framing & Theory Notes

1 What is Regression? How is it different from Classification? Give 2 real-world Indian examples.

Answer:

- **Regression** predicts a continuous number (price, amount, yield). **Classification** predicts a category or label (yes/no, low/mid/high).
- Regression is judged by error (RMSE, MAE, R^2); classification by accuracy, precision and recall.
- **Example 1:** predicting the EMI amount at HDFC from loan amount, tenure and interest rate.
- **Example 2:** predicting crop yield (quintal per acre) in Maharashtra from rainfall, soil and fertiliser use.

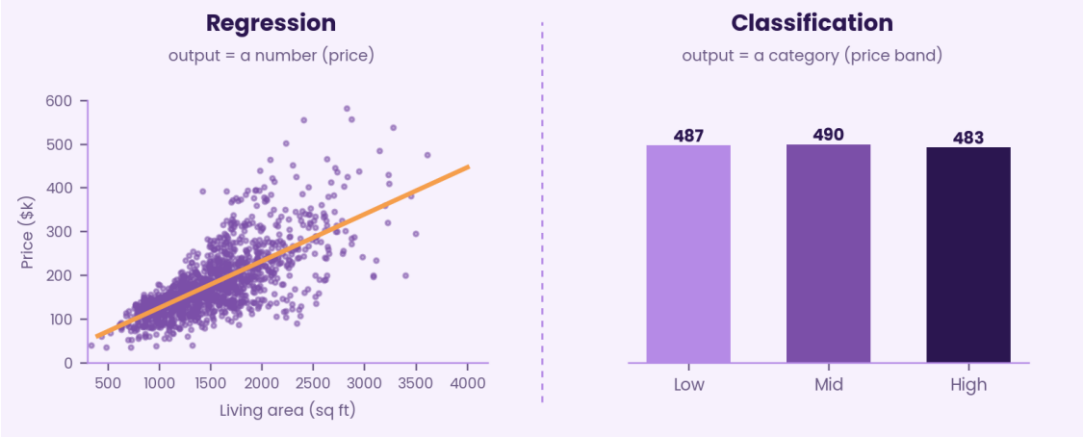


Figure 1: Our task (sale price in $) is regression. Cutting the same prices into 3 bands would make it classification.

2 What is the difference between Simple Linear Regression, Multiple Linear Regression, and Regularised Regression (Ridge / Lasso)?

Answer:

- **Simple:** one feature, one straight line, y = b0 + b1·x (e.g. SalePrice from GrLivArea).
- **Multiple:** many features together, y = b0 + b1·x1 + … + bn·xn (area, quality, garage, etc.).
- **Regularised:** multiple regression plus a penalty on coefficient size. **Ridge (L2)** shrinks coefficients; **Lasso (L1)** can push weak ones to exactly zero.

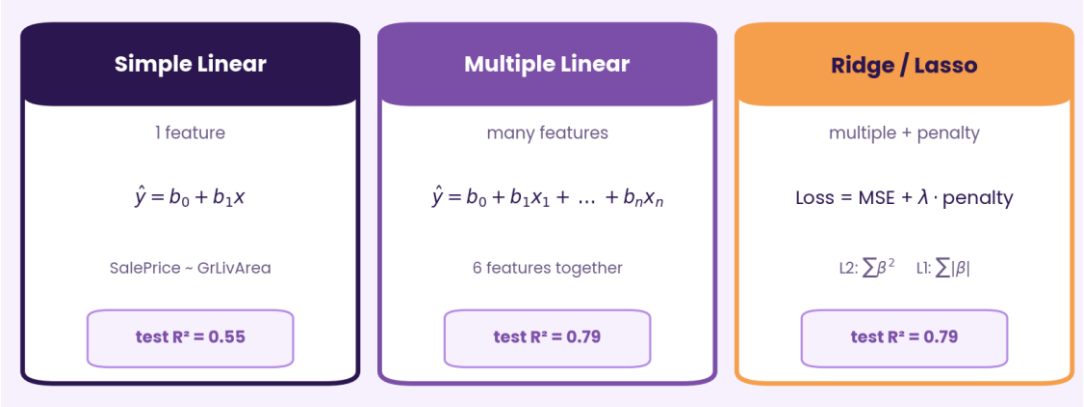


Figure 2: The three models side by side, with test R^2 on our 6-feature Ames subset (Ridge/Lasso gain is small here as there are few features).

3 Explain Overfitting and Underfitting. How does regularisation (L1/L2) help prevent them?

Answer:

- **Underfitting:** model too simple, high error on both train and test data (high bias).
- **Overfitting:** model memorises noise, very low train error but high test error (high variance).
- **Regularisation** adds λ × (size of coefficients) to the loss, so the model cannot chase noise. L2 shrinks all coefficients smoothly; L1 drops weak features. Bigger λ gives a simpler model.

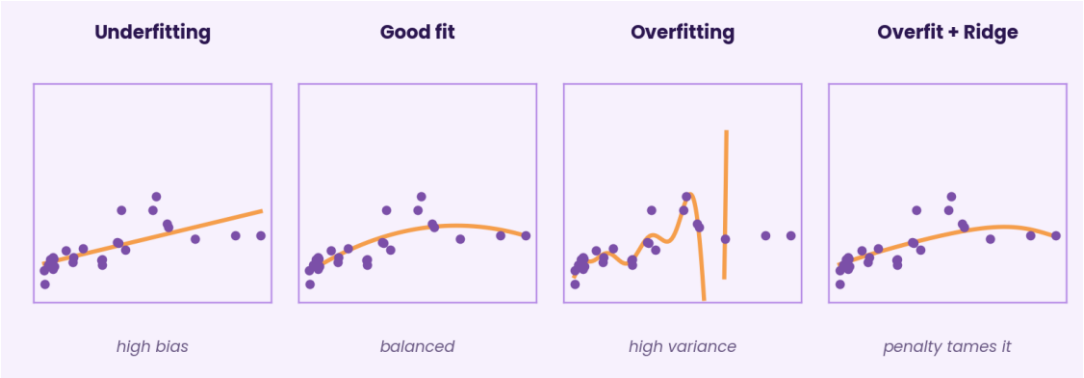


Figure 3: Polynomial fits on 28 random Ames houses. Degree 12 chases noise; a Ridge penalty smooths it again.

4 Define RMSE, MAE, and R^2. When would you prefer RMSE over MAE?

Answer:

- **MAE** = mean of |actual − predicted|: the average error, in the same unit as price.
- **RMSE** = √mean((actual − predicted)²): errors are squared first, so big misses hurt more.
- **R^2** = 1 − SSres / SStot: share of price variation the model explains (1 is perfect, 0 is no better than the mean).
- Prefer **RMSE** when large errors are especially costly (badly mispricing an expensive house). Prefer MAE when a few outliers should not dominate.

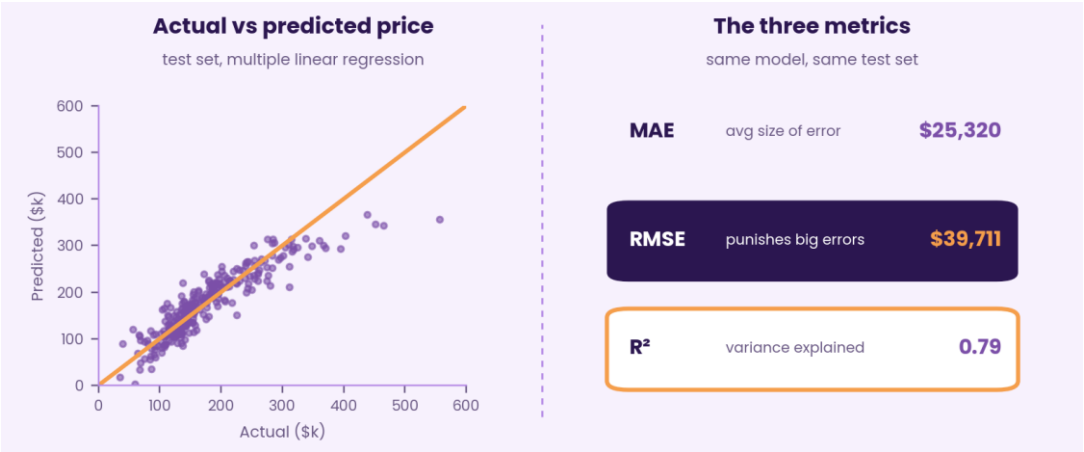


Figure 4: On the test set our model gets MAE ≈ $25,320, RMSE ≈ $39,711, R^2 ≈ 0.79. RMSE > MAE because of the big misses.

5 What is Cross-Validation (k-Fold)? Why is it preferred over a single train-test split?

Answer:

- Split data into **k equal folds**. Train on k−1 folds, test on the remaining one, repeat k times, then average the k scores.
- A single split can be lucky or unlucky. k-fold uses every row for both training and testing, so the score is steadier.
- It also shows the spread (± std) and is used to tune hyper-parameters like λ, which matters with only 1,460 rows.

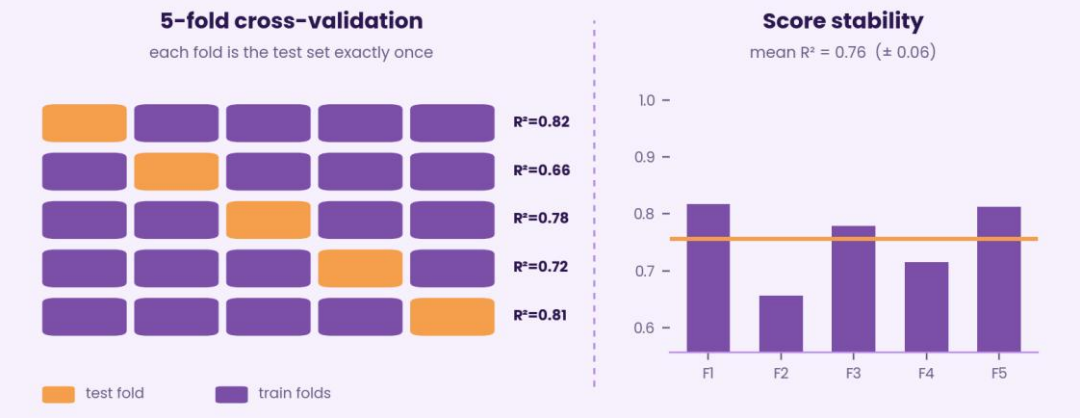


Figure 5: 5-fold CV on our 6-feature model gives R^2 from 0.66 to 0.82. A single split would show only one of these.

Quick Recap

Regression	Predicts a continuous number (price); classification predicts a category.
Simple → Multiple → Ridge/Lasso	One feature → many features → many features + coefficient penalty.
Over/Underfitting	Too complex memorises noise; too simple misses the pattern. Regularisation controls it.
MAE · RMSE · R^2	Average error · error that punishes big misses · share of variance explained.
k-Fold CV	Average of k test scores: a steadier estimate than one train-test split.